# COLLAPSING A FIBRE OF A PROPER FAMILY ONTO A SEPARATED FINITE-TYPE TARGET FORCES THE WHOLE FAMILY NEAR THE BASE POINT TO FACTOR THROUGH A FINITE SUBSPACE

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*Editorial note.* Statement span is lines 7930-7958 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 7960-7978, the complete author proof (17 body lines).

*Editorial note.* Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Statement carries the author's \begin{slogan} block verbatim in the retained span.

*Editorial note.* Counted B side; schemes twin uncounted per ruling.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. Morphisms of Spaces, sections/lemmas scheme-theoretic-image and scheme-theoretic-image-is-proper; Decent Spaces, lemma conditions-on-fibre-and-qf; in-source lemma-proper-finite-fibre-finite-in-neighbourhood (card L16242).

*Source reference.* Original lemma 0ACK: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7978–7978. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 02UP: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7906–7913. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 0AGD: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7978–7978. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 082W: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 7978–7978. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

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*Date*: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

**Lemma 0.1.** *Collapsing a fibre of a proper family forces nearby ones to collapse too. Let $S$ be a scheme. Let*

$$\begin{array}{ccc} X & \xrightarrow{h} & Y \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle g} & \\ & B & \end{array}$$

*be a commutative diagram of morphism of algebraic spaces over $S$. Let $b \in B$ and let $\mathrm{Spec}(k) \to B$ be a morphism in the equivalence class of $b$. Assume*

(1) *$X \to B$ is a proper morphism,*
(2) *$Y \to B$ is separated and locally of finite type,*
(3) *one of the following is true*
   (a) *the image of $|X_k| \to |Y_k|$ is finite,*
   (b) *the image of $|f|^{-1}(\{b\})$ in $|Y|$ is finite and $B$ is decent.*

*Then there is an open subspace $B' \subset B$ containing $b$ such that $X_{B'} \to Y_{B'}$ factors through a closed subspace $Z \subset Y_{B'}$ finite over $B'$.*

*Proof.* Let $Z \subset Y$ be the scheme theoretic image of $h$, see Morphisms of Spaces, Section 082W. By Morphisms of Spaces, Lemma 0AGD the morphism $X \to Z$ is surjective and $Z \to B$ is proper. Thus

$$\{x \in |X| \text{ lying over } b\} \to \{z \in |Z| \text{ lying over } b\}$$

and $|X_k| \to |Z_k|$ are surjective. We see that either (3)(a) or (3)(b) imply that $Z \to B$ is quasi-finite all points of $|Z|$ lying over $b$ by Decent Spaces, Lemma 0ACK. Hence $Z \to B$ is finite in an open neighbourhood of $b$ by Lemma 02UP. □